

A JXTA stack

JXTA is a protocol, like TCP/IP or FTP, being developed by Sun Microsystems, which could be used to develop a distributed network using P2P technology, and to develop basic building blocks and services for P2P networks.

The present P2P applications, excluding Groove and to a certain extent Gnutella, are capable of performing only one task. For example, Napster is only for file-sharing, SETI@Home or Rumor only for resource sharing, and so on. What JXTA plans to achieve is to design an application that will be able to handle file-sharing, instant messaging, as well as resource sharing from one single interface.

application based on a P2P architecture whose goal is to connect people over the Internet or a local network and then allow them to work together on a common project.

The software allows users to collaborate in encrypted, shared spaces that are stored on the computer of each member of the virtual meeting place. This can be beneficial for enterprises because it lets them provide employees with an environment that is conducive to collaboration while also utilising the storage capacity of the hard drives of users instead of having to dedicate a central file server to the task. The ability to update files when offline clients log on and optimum bandwidth utilisation are other reasons why Groove is useful for individuals who do not have high Internet or LAN speeds.

Consider a Groove-generated collaborative environment where three people are working on the same file. Person A makes a change to the file, which means that the remaining two should

Sample Gnutella Network



Rumor Technology is used to broadcast antivirus and firewall configuration updates to all computers on a single network

be informed about the change. Instead of sending the complete file, Groove sends only the delta (a data packet in Groove) for the changed portion, thus saving bandwidth.

OpenCola Folders is another product that attempts to tackle the problem of information overload caused by inefficiencies in conventional search engine technologies. It allows users to configure the software to search for information with the option to specify additional parameters such as document type. The software then goes out and checks whether other OpenCola Folders can provide it with the required information.

So, what's next?

The fact that P2P technology provides developers with a networking architecture that facilitates direct communication between people makes it very popular for applications that involve the exchange of data. However, this is just the tip of the iceberg.

Coming soon are intelligent search engines like BadBlue (www.badblue.com) that use P2P agents to search networks for files that have been publicly shared by users. This will probably lead to the next step in the evolution of search engine technology.

But some of the more complex applications that are beginning to come out now are those that improve the utilisation of network resources. Chaincasting, from Chain-Cast Networks (www.chaincast.com/technology/chaincasting.html), involves the use of P2P technologies to establish streaming chains that reduce the costs associated with the high bandwidth requirements for video streaming.

Whether it is entertainment or work, P2P technology is not just making its presence felt, but is also providing a viable and preferred alternative to client-server networking. So if you still aren't a link in the P2P network, prepare to become one very soon. ■

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